

SAYANI MALLICK

9453422929 • sayanimallick0218@gmail.com • linkedin.com/in/sayani-mallick-80825018b/ • github.com/sayanim1

EDUCATION

MS, Computer Science

University of Texas, Dallas
Richardson, USA

Aug 2025 - May 2027
4.0/4.0 GPA

Relevant coursework: Artificial Intelligence, Database Design, Computer Vision, Natural Language Processing

B.E., Electronics and Communication Engineering

BITS Pilani, KK Birla Goa Campus
Goa, India

Aug 2019 - Jul 2023
8.3 GPA

Relevant coursework: DSA, Object Oriented Programming, Neural Networks, Probability and Statistics

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, C++ , HTML , ReactJS, VueJS, CSS, NodeJS

Data & Backend: Distributed Systems, SQL, MongoDB, REST APIs, Kafka, Spark, AWS, Elasticsearch, Multithreading, Kubernetes, Databases, Microservices, FastAPI

AI / ML: Machine Learning, OpenCV, PyTorch, TensorFlow, Scikit-learn, Numpy, Image Processing, Time Series Modeling, Model Evaluation, Large Language Models, HuggingFace, Vector databases(Chroma, FAISS) , Prompt engineering, LLMs, RAG, LangChain, API integration, Model training

Tools: Git/GitHub, JSON, Jenkins, CI/CD, Linux, Shell scripting, Grafana, A/B testing, Docker

EXPERIENCE

Amazon, Seattle, Washington: Software Development Engineer intern

Jun 2026 - Aug 2026

- Engineered an end-to-end contextual AI search experience within the Prime Video mobile player using Kotlin, AWS Lambda, REST APIs, and Amazon Bedrock, leveraging playback context to deliver personalized, context-aware recommendations; optimized search relevance and ranking and implemented seamless content launch from search results, achieving 2.5s P90 latency for the end-to-end search API
- Engineered and deployed the end-to-end solution, leveraging **Jetpack Compose + UDF, AWS Lambda**, parallel catalog validation, and **AWS CDK** to deliver a scalable, production-ready architecture.

Flipkart, Bangalore: Software Development Engineer - I and intern

Jul 2023 - Aug 2025

- Designed a priority-based ingestion scheduler with MongoDB and Kafka Streams, cutting product refresh latency from 12 hours to 2 hours and boosting throughput and SLA compliance.
- Engineered a Kafka–Spark–Hadoop (MapReduce + Java) pipeline to compute new vs. repeat customer metrics, powering the Brand Insights Tool and driving a 20% increase in new brand listings.

PROJECTS

Machine Learning-Led Closed Loop Adaptation of Exergaming – HeXD Lab, UTD

Designed a deep-learning powered exergame agent that adapts dance difficulty in **real time** by integrating ECG signals, **computer vision** pose estimation, and game state under the guidance of Prof. Rawan Alghofaili.

Built Python data pipelines to ingest and preprocess physiological and pose data and trained ML models to estimate exertion and fatigue, driving automated difficulty adjustments in a closed-loop, agent-style control system.

TrustIssues.AI – Multi-Agent Compliance & Onboarding Assistant (HackUTD 2025 Winner)

Built a Nemotron-powered multi-agent system with five specialized agents to automate vendor onboarding via document parsing, registry validation, risk scoring, and explainable access recommendations, reducing onboarding time and improving compliance transparency.

Long-term Memorability for Advertisements (Collaboration: Cognitive Neuroscience Lab and Adobe Research India) : Designed and conducted experiments for understanding correlation between long-term memorability of advertisements and eye-saccades, fixation of viewers. Developed saliency maps for understanding attention distribution.

Novel Deep Learning Framework for Detection of Epileptic Seizures [Paper]

Developed novel 1D-CNN + Bi-LSTM architecture for epileptic seizure detection, achieving 98.03% accuracy on the Bonn EEG dataset; demonstrated expertise in time-series modeling and deep learning for biomedical signals